MIL-E-1/999A
13 June 1972
SUPERSEDING
MIL-E-1/999(SigC)
23 November 1955

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6163

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: ATR, broadband, Fo = 9,050 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	TA (nonoperating)	Alt
Unit:	kw	---	°C	ft
Maximum:	250	0.001	+100	10,000
Minimum:	4	---	-40	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Mounting position:	Any
Test mounts:	See figures 2, 3, and 4
Weight:	0.13 pound (approx)
Application:	See note 6

TEST CONDITIONS:

Parameter:	Incident power	tp	prr	σ'	F
Unit:	kw	μs	pps	---	MHz
Tolerance:	±5%	±0.15	±10%	Max	---
Test condition 1:	4	0.5	1,000	---	F4
Test condition 2:	20	1.0	1,000	1.02	F3
Test condition 3:	200 (min)	1.0	1,000	1.2	F4

Frequency		
F	MHz	±
1	9,050	---
2	9,050	0.1%
3	9,050	1.0%
4	9,050	3%

GENERAL:

Preproduction sample approval - Required (see note 1)

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Preproduction sample approval inspection	1	-					
4461	Loaded Q	2	-	Fo = F1	QL	---	6.5	---
4474	High-level VSWR	2	2		VSWR	---	1.15	---
4021	Degradation due to vibration	-	-		---	---	---	---
	Quality conformance inspection, part 1	3	-					
4482	Tuning susceptance	2	-	Fo = F2; g=0.05	b	-0.06	+0.06	---
4486	Firing time	4	1		t	---	10	sec
4488	Arc loss	4	1		La	---	0.8	dB
---	Electrical symmetry	2, 5	-	Fo = F2; g=0.05	---	---	0.06	cm
1027	Temperature cycling (nonoperating)	-	-		---	---	---	---
	Quality conformance inspection, part 2							
4101	Dielectric material strain	-	-		---	---	---	---
4484	Normalized conductance	2	-	Fo = F2	g	---	0.06	---
4471	Recovery time	2	3		t	---	10	μs
	Quality conformance inspection, part 3							
---	Life test	7	3	Group D	t	500	---	hrs
---	Life-test end points:							
4482	Tuning susceptance	2	-	Method A; Fo = F2; g = 0.05	b	-0.08	+0.08	---
4488	Arc loss	4	1		La	---	1.0	dB
4484	Normalized conductance	2	-	Fo = F2	g	---	0.1	---
4471	Recovery time	2	3		t	---	15	μs
1027	Temperature cycling life test	-	-	Group D; 10 cycles (min)	---	---	---	---

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NOTES:

1. Preproduction sample approval requirements hereby replace any qualification requirements referable to the product covered herein. The term "First article testing" shall be considered as synonomous with the term "Preproduction sample approval testing." All tests applicable herein (including all preproduction sample approval and quality conformance inspection, parts 1, 2, and 3) shall be performed during preproduction sample inspection. A failure of any one tube in any of the tests shall be cause for decision of preproduction sample disapproval.
2. The TUT shall be mounted in test mount A as detailed in figures 2 and 3.
3. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
4. This test shall be performed at least 168 hours after pumping and at least 24 hours after any previous discharge.
5. The shift in position of the voltage minimum, resulting from reversing the TUT in the test mount, shall be within the specified limits when measured in a RG-51/U waveguide.
6. This tube is designed for application in pairs mounted in the wide walls of RG-51/U waveguide.
7. The TUT shall be mounted in test mount B as detailed in figure 4.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - NS
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2575)

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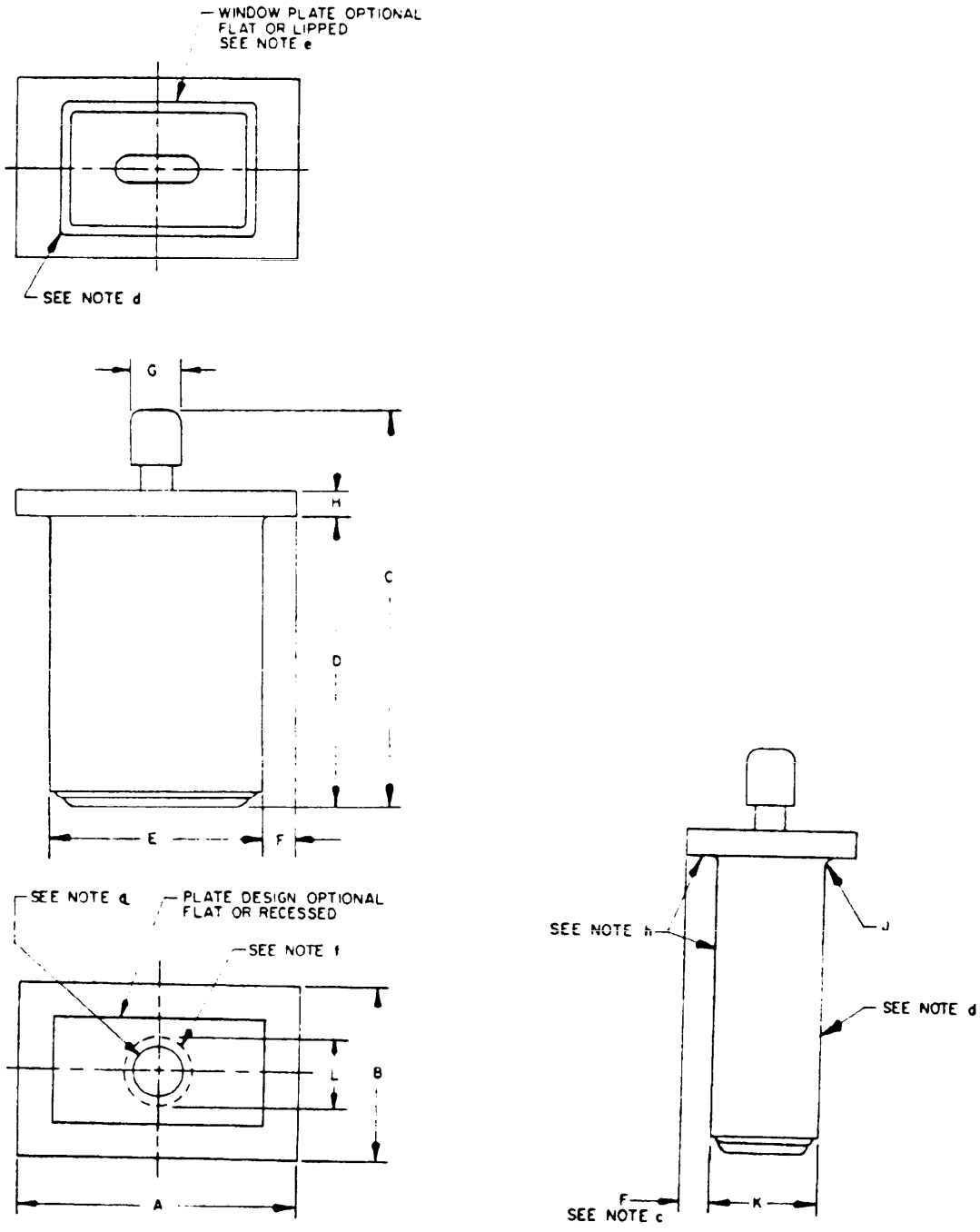


FIGURE 1. Outline drawing of electron tube type 6163.

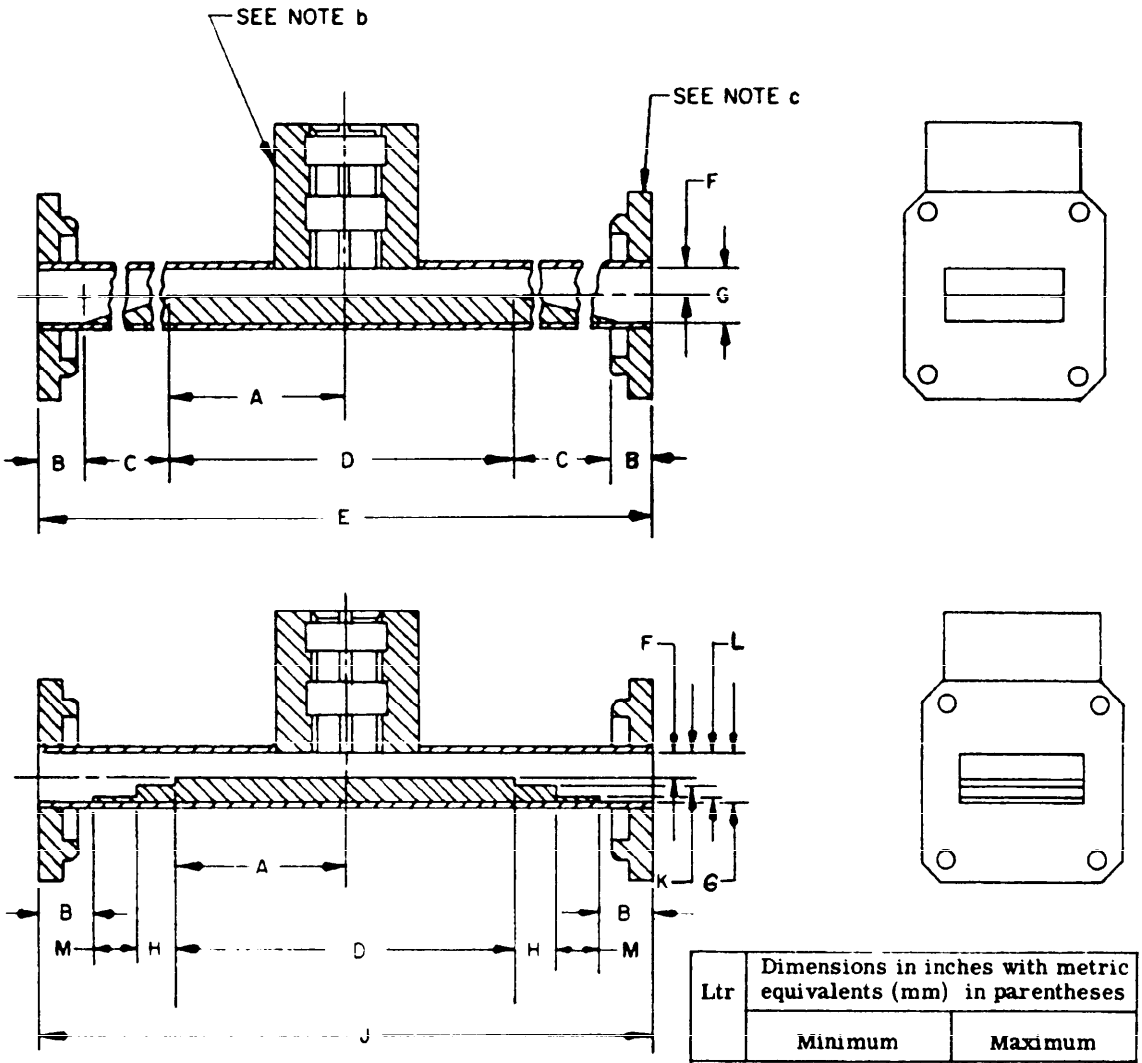
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Preproduction sample approval inspection (See note 1)		
G		.281 (7.14)
Quality conformance inspection, part 1 (see note 3)		
A	1.265(32.13)	1.285(32.64)
B	.770(19.56)	.780(19.81)
C		1.719(43.66)
D	1.296(32.92)	1.302(33.07)
F	.125 (3.18)	
Quality conformance inspection, part 2		
E	.995(25.27)	1.005(25.53)
H	.117 (2.97)	.133 (3.38)
J		.020RAD(.51)
K	.495(12.57)	.505(12.83)
L		.438(11.13)

NOTES:
a. The tubulation shall fall within a circle .438(11.13 mm) dia max located from the center lines of the end plate.
b. Silver plate 100MSI min or equivalent finish.
c. Applies to four walls for full length from radius J to window plate.
d. Four long edges of tube body shall have a .020(.51 mm) ±.010(.25 mm) approx radius.
e. A radius of .031(.79 mm) max or a chamfer of .016 (.41 mm) X 45° max permissable on the window plate corners.
f. Spread of solder to be held within area shown.
g. No gaskets are to be used with this tube.
h. These surfaces on all four sides shall be perpendicular to each other within .010(.25 mm) for full length of body from radius J to window plate.

FIGURE 1. Outline drawing of electron tube type 6163 - Continued.

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ALTERNATE DESIGN OF TEST MOUNT

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection		
A	1.590(40.39)	1.610(40.89)
C	3.190(81.03)	3.210(81.53)
D	3.190(81.03)	3.210(81.53)
F	.248 (6.30)	.250 (6.35)
G	.494(12.55)	.500(12.70)
H	.396(10.06)	.406(10.31)
K	.309 (7.85)	.313 (7.95)
L	.371 (9.42)	.375 (9.53)
M	.394 (10.01)	.414 (10.52)
Reference dimensions		
B	.50 (12.70)	
E	10.59(268.99)	
J	5.81(147.57)	

NOTES:
a. Method of fabrication optional.
b. Dimensions of socket shown on figure 3.
c. Flanges may be UG-51/U or UG-52 A/U.
d. All internal edges to have .015(.38 mm) to .031R(.79 mm).

FIGURE 2. Test mount "A" outline.

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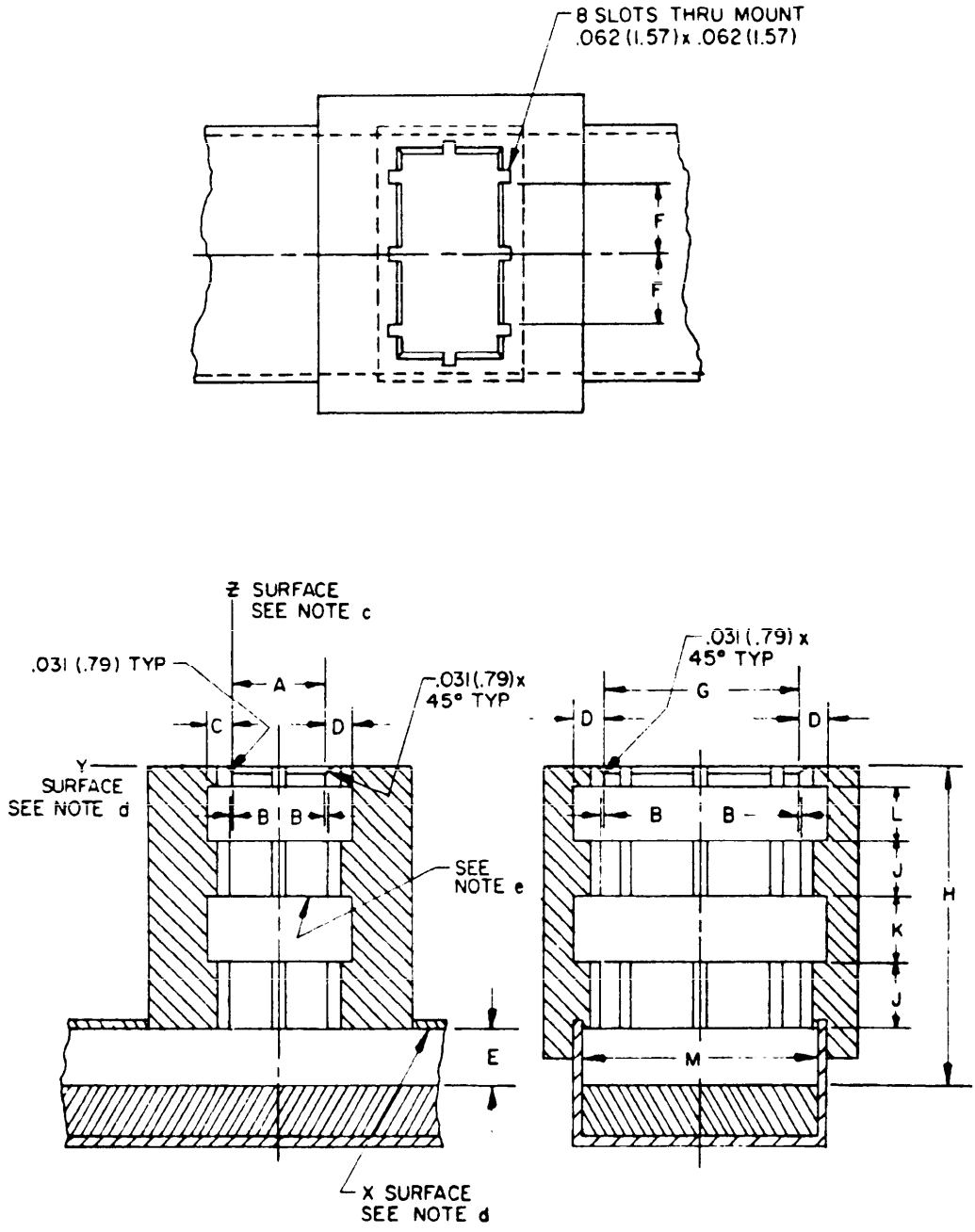


FIGURE 3. Test mount "A" details.

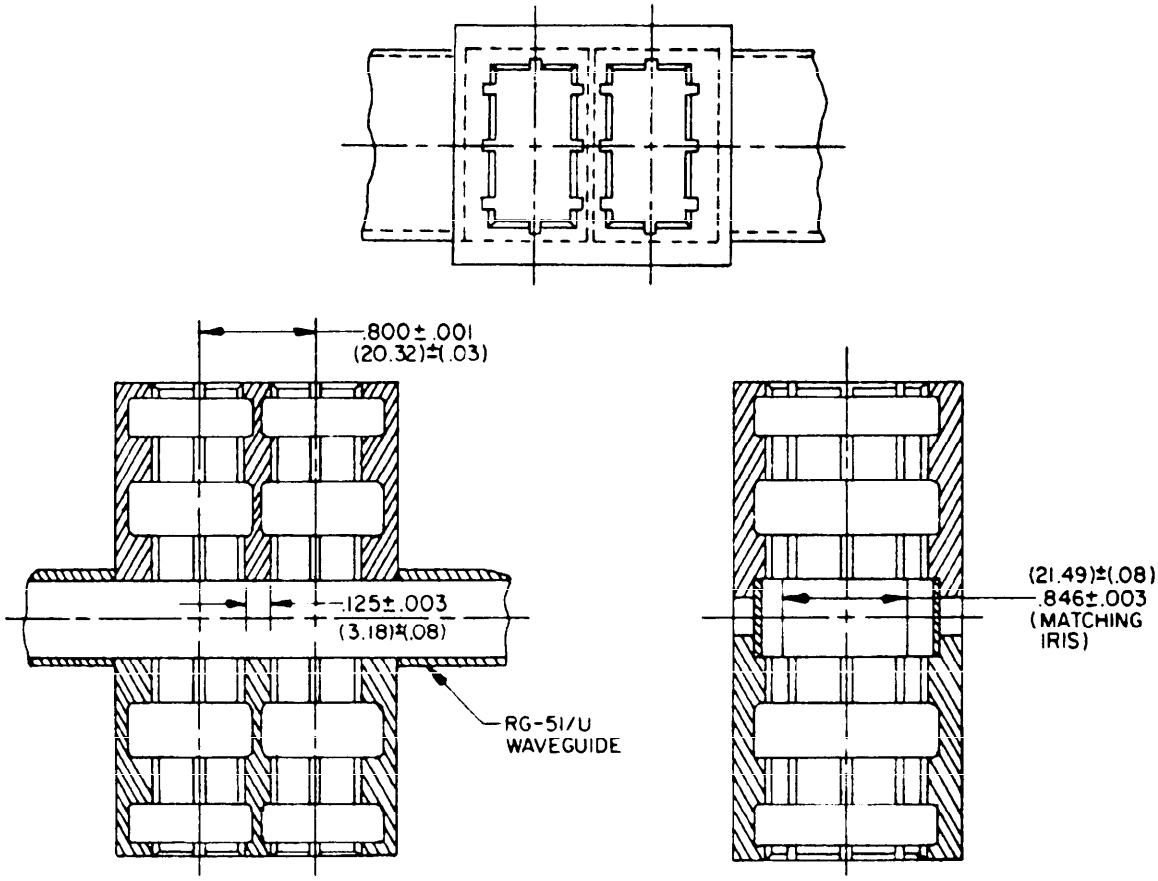
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection		
A	.506(12.85)	.511(12.98)
B	.016 (.41)	.020 (.51)
C	.126 (3.20)	.135 (3.43)
D	.196 (4.98)	.205 (5.21)
G	1.006(25.56)	1.012(25.70)
H	1.550(39.37)	1.554(39.47)
J	.288 (7.32)	.298 (7.57)
K	.352 (8.94)	.362 (9.19)
L	.257 (6.53)	.267 (6.78)
Reference dimensions		
E	.249 (6.32)	
F	.344 (8.74)	
M	1.122(28.50)	

NOTES:
a. Method of fabrication optional.
b. Suitable clamp shall be provided to insure proper seating of tube in mount.
c. Pressure shall be provided to press the tube against surface "Z".
d. Surfaces "X" and "Y" shall be parallel within $\pm 0^{\circ} 5'$.
e. All internal edges to have .015R(.38 mm) to .031R(.79 mm).

FIGURE 3. Test mount "A" details - Continued.

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NOTES:
a. Internal dimensions of each socket same as detail of test mount "A".
b. Method of fabrication optional.
c. Suitable clamps shall be provided to insure proper contact of tubes in mount.

FIGURE 4. Test mount B.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/999A ELECTRON TUBE, GAS SWITCHING TYPE 6163

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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